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Financial literacy and sustainable planning assessment among Filipino millennials

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ABSTRACT

Financial literacy plays an important role in improving personal financial management and helping individuals reach their financial goals, especially for millennials, who currently represent the largest working-age demographic and key consumers. This study sought to address the gaps in low financial literacy levels and poor financial decision-making among Filipinos, as well as the lack of publicly available studies on millennial financial literacy in the Philippines particularly in relation to financial planning. The study assessed the financial literacy of Filipino millennials and its influence on financial planning by employing a correlational research method to test four hypotheses using data from 400 respondents. Financial literacy was measured through "The Big Three" questions, and structural equation modeling was applied to analyze the relationships between financial literacy and demographic factors to financial planning, with goal-setting moderators (goal commitment, goal specificity, goal acceptance, and goal difficulty) serving as the basis for financial planning. Results indicated low levels of financial literacy – leading to the insignificant relationship on sustainable planning, while demographic factors such as monthly income, occupation, and monthly expenses showed high significant influence on financial planning. This study provided additional understanding on the financial literacy and planning of Filipino millennials, offered implications and financial sustainable strategies for the government, financial institutions, and educational institutions aiming to improve financial literacy and planning. Additionally, it emphasized the need for more targeted financial inclusion initiatives tailored to Filipino millennials.

1. Introduction

Financial behavior encompasses a wide range of decisions, including saving, spending, investing, borrowing, and budgeting. It involves developing strategies for informed financial choices and adopting sustainable habits to manage resources effectively and achieve financial goals (Aryan et al., 2024; Fan, 2020). Financial literacy is the understanding of financial management knowledge and skills, enabling one to regulate investments and manage money to reach financial goals (Alshami et al., 2024; Tiwari & Yadav, 2022). It was suggested that financial behavior is linked to how individuals use and allocate their financial resources, and the positive action pertains to financial literacy (Bhargava et al., 2022). Similarly, Angrisani et al. (2020) discussed the importance of financial decisions on a path towards financial security, where those with greater financial behavior and literacy are better equipped to handle debt and have a strong predictive ability for future financial outcomes.

Surveys like the Financial Literacy Around the World (FLAT World) project, National Financial Capability Study (NFCS), Survey of Consumer Finances (SCF), and National Financial Educators Council (NFEC) Financial Literacy Survey, along with various research papers (Arceo-Gómez & Villagómez, 2017; Mohta & Shunmugasundaram, 2022; Sconti, 2022; Smolo et al., 2023), highlighted the global importance of financial literacy. In the 2014 Standard & Poor's Financial Literacy

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Survey, more than 140 countries - including the Philippines - were included in the study. It was revealed that only 25 % of the adult population in the Philippines demonstrated financial literacy, placing the country 13th among 21 developing Asian countries. Based on the report, emphasis is needed on the importance for improved financial literacy in the Philippines.

Grohmann (2018) studied financial behavior and literacy in emerging Asia, focusing on over 500 middle-class individuals in Bangkok. Despite the rapid growth of the financial sector, which required the middle class to quickly adapt to new financial products, only 24 % of respondents had advanced financial knowledge. The study revealed a common relationship between socio-demographic variables and financial literacy, with wealthier individuals and those with higher incomes showing greater financial literacy (Grohmann, 2018). This pattern closely mirrors the situation in the Philippines, where financial literacy levels are also low. In accordance, targeted efforts are needed to enhance financial education, particularly among socio-demographic groups with lower incomes and education levels.

In the current generation, millennials have recently dominated the working-age population. It indicates that these individuals are most likely using financial services and are seen as a consuming generation (Dewi et al., 2020). In context, financial literacy is said to be influenced by both internal and external factors, such as cognitive abilities and the state of the economy (Fridayani & Sadewo, 2018). It is important to consider the level of financial literacy because it symbolizes the way in which individuals establish their financial goals and make choices to achieve financial well-being (Soekarno & Pranoto, 2020). Aryan et al. (2024) proposed that decisions regarding the allocation, use, and management of financial resources are related to people's financial behavior. In terms of millennial investors, the report indicated that millennials are categorized under the 'low-advanced financial literacy' criteria. The report suggests that millennials' level of financial literacy is not high enough (Mohta & Shunmugasundaram, 2022). According to Tiwari and Yadav (2022), financial behavior plays a crucial role in making decisions in any area of an individual's life and understanding the financial products that most people interact with is known as financial literacy. Thus, the need to understand financial literacy in relation to financial planning is crucial, which is especially important since millennials are currently considered to be the adult age group.

The Consumer Financial Survey conducted by the Banko Sentral ng Pilipinas (BSP) (Banko Sentral ng Pilipinas, 2018) revealed that the financial assets and spending patterns of Filipino households. The survey found that only one in four Filipino households had access to financial products and services. Specifically, 16.3 % of households had insurance or pension benefits, while 9.7 % held deposits. Moreover, less than 1 % owned other financial assets, such as e-money, with the persistent lack of funds cited since the 2014 survey as the main reason for not owning these assets. On the other hand, nearly half (49.9 %) of household spending is dedicated to food and non-alcoholic beverages, with 23.9 % going to housing and utilities. Households also spend more on alcoholic beverages, tobacco, and narcotics (5.8 %) than on health (4.8 %), despite health being essential.

Addressing these discrepancies is necessary, as better financial management is linked to improved financial access (Khan, Siddiqui, & Imtiaz, 2022). The financial well-being of individuals and households is a crucial concept in research on financial literacy, competence, capability, education, and behavior (Oberrauch et al., 2023; Van Raaij et al., 2023). Therefore, promoting effective personal financial management will help Filipinos achieve their goals through budgeting, saving, investing, and debt management (Eloriaga et al., 2022). This, in turn, will support financial development and positively impact well-being (Pham et al., 2018).

In the study by Mohta and Shunmugasundaram (2022), it was stated that there is a significant relationship between an individual's income and their level of financial literacy, where higher earners exhibited higher levels of financial literacy. Studies that were conducted showed that Filipinos' primary concerns are poor financial decisions and income loss, which makes them victims of investment schemes and scams (Galapon & Bool, 2022). According to an article by the Philippine News Agency by Ledesma (2021), a survey showed that the Philippines has the lowest in Asia who set aside for retirement. Subsequently, most of the population learned about investing much later in life or knew very little about it at all. It represents how millennials' negative financial behavior includes things like accumulating debt, engaging in risky or speculative investments without enough knowledge or understanding, living above one's means, and spending excessively (Fan, 2020). In addition, Walczak and Pieńkowska-Kamieniecka (2018) argued that there are differences in financial behavior and attitude in using banking services and other financial instruments – leading to difficulty in assessing overall financial literacy, stability, sustainability, and behavior. To which it should be assessed in order to understand the capability of Filipino millennials on financial sustainability. In relation to relevant studies, the measurement of financial literacy has been benchmarked for holistic assessment.

The goal setting theory demonstrates broad applicability across various research fields due to its frequent use in studies exploring relevant topics such as physical activity and sports (Jeong et al., 2023; Swann et al., 2020), pediatric rehabilitation (Pritchard-Wiart et al., 2019) and in organizational practice (Ogbeiwi, 2022). To date, it has shown generality across participants, tasks, nationality, goal source, settings, experimental designs, outcome variables, levels of analysis (individual, group, division, and/or organizational), and time spans (Locke & Latham, 2019). A review of online scholarly databases revealed a scarcity of recent financial research employing goal setting theory. The Organisation for Economic Co-operation and Development (OECD) large scale measurement of financial literacy has contributed to advancements in the field, as the earliest studies that began in the 1960s and 1970s were only limited to money management practices (Sudakova, 2018). The application of theory to financial research has been deemed appropriate and important because it provides a holistic assessment of financial literacy (Mu'izzuddin et al., 2017).

Notably, the most recent study identified predates 2018, suggesting a potential gap in the application of the theory within the financial domain. Therefore, it could be posited that a full-on understanding of financial literacy and financial planning could be comprehensively assessed with the integration of "The Big Three" questions by the Global Financial Literacy Excellence Center (GFLEC) and goal setting theory. In relation, financial literacy's effect on Filipino millennials' financial planning remains under-investigated, and the application of the Goal Setting Theory developed by Locke and Latham (1990) as a research framework within the field of finance is underrepresented in scholarly literature, as a dearth of readily available studies specifically targets this demographic and theoretical framework.

To date, there have been no studies in the Philippines that covered the financial literacy and planning among Filipinos despite the crucial steady rise of economic problems in the country. Nonetheless, this is evident in other Asian Countries like China from Chen and Chen (2023) and India from Agarwal et al. (2015). These studies have presented how financial literacy in a subjective and objective point of view have affected retirement plans among Chinese, and higher financial literacy related to highly significant investment behavior and financial planning among Indians. Broader review also provided evidence that higher financial literacy leads to higher and better retirement plans in the review study conducted by Zaimovic et al. (2023) from people in the United States, Germany, Finland, Chile, Sweden, China, Canada, Ghana, New Zealand, South Africa, and Russia, among other developed countries. Utilizing a novel framework that was originally made to holistically assess financial planning, a comprehensive analysis of financial literacy from characteristics and behavior of millennials as the current adults in the country was conducted. Considering the inter- and causal relationships of the variables in this study, the partial least squares structural equation modeling was deemed appropriate following related studies (Dash & Paul, 2021; Hair et al., 2021).

The purpose of this study was to determine the financial literacy of Filipino millennials and its effect on financial planning with the integration of goal setting theory and "The Big Three" GFLEC question. Specifically, the assessment of financial literacy was employed through three questions on interest compounding, inflation, and risk diversification (Lusardi & Mitchell, 2011), while sociodemographic characteristics such as monthly income, occupation, and monthly expenses were utilized to holistically analyze their effect on financial planning. The results of this study hold a significant impact on knowledge acquisition and the continued development of the financial system with beneficiaries including millennials, financial and non-financial institutions, the Philippine government, policymakers, future scholars, and future generations. Theoretical, practical, and managerial implications were suggested for actual practice based on the findings of this study.

The current study presents the introduction as section (1), which covered the background of the study, practical and theoretical problems, related studies and research gap, research need and originality, and the study objective and significance. This was followed by the discussion of conceptual framework, build up, and hypotheses in section (2). Section (3) and section (4) covered the methodology and results, respectively. The participants and data collection, questionnaire, tool, and output were reported. The succeeding sections covered the discussion of results, practical and theoretical implications, strategies, and study limitations and future direction; alongside the study conclusions in sections (5, 6), respectively.

2. Conceptual framework

The conceptual framework for this research is shown in Fig. 1, illustrating the direct effect of financial literacy on financial planning – reflecting a higher-order construct of goal commitment, goal specificity, goal acceptance, and goal difficulty adapted from Locke and Latham (1990). Moreover, lower-ordered constructs such as monthly income, occupation, and monthly expenses were indicated to affect financial planning. A total of four (4) hypotheses were considered to assess its direct effect on financial planning among people from the Philippines.

The "Big Three" financial literacy questions by Lusardi and Mitchell (2011) – which covered the concepts of interest compounding, inflation, and risk diversification (Lusardi, 2019) – were utilized in this study to measure financial literacy. The set of questions has been widely adopted in numerous surveys and studies around the world (Mohta & Shunmugasundaram, 2022; Sconti, 2022; Smolo et al., 2023), demonstrating its effectiveness as a standardized measure of financial literacy. It is said that financial literacy is achieved when the majority of respondents obtain a passing grade of 2 out of 3 from the "Big Three" questions as it covers three vital economic concepts essential for individuals making financial decisions. Related studies (Arsyadanie & Rahadi, 2024; Culebro-Martínez et al., 2024; Khyareh & Zivari, 2023; Lusardi et al., 2020; Oehler & Horn, 2023) have explained each domain being a significant determinant of financial literacy which demonstrates that interest compounding, inflation, and risk diversification contributes to financial literacy. These therefore serve as reflective constructs of financial literacy applied in this study.

Compounding interest can create a snowball effect that makes money grow faster (Arsyadanie & Rahadi, 2024). Individuals with higher financial literacy tend to be more inclined to plan for retirement, probably due to their understanding of the impact of compound interest and their proficiency in performing calculations (Lusardi, 2019). The individual measures of financial literacy across countries show that being literate in interest rates and understanding risk diversification are positively associated with savings (Beckmann & Kiesl-Reiter, 2023). Also, the "Big Three" shows that it is particularly important to make people familiar with the concepts of risk and risk diversification (Lusardi, 2019). Jumena et al. (2022) defined inflation as the reflection of the rate at which prices of goods consumed by the public change, and from the viewpoint of currency exchange rates with goods and services, it signifies the weakening of money in exchange for purchased items.

In Eastern European countries, financial literacy (including

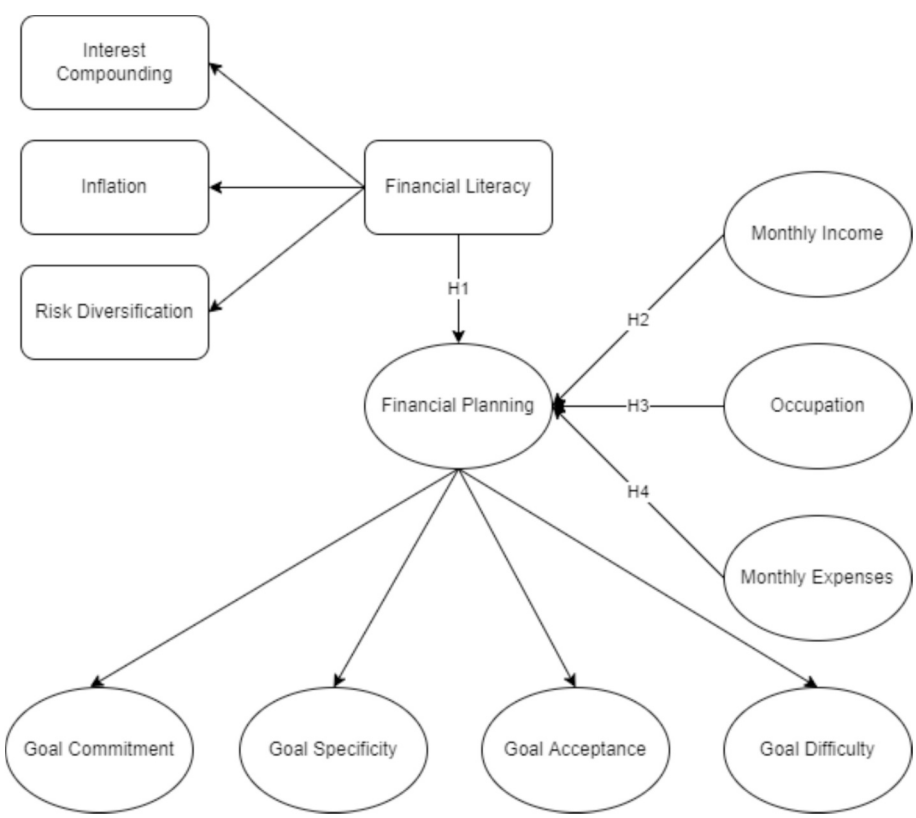

Fig. 1. Conceptual framework.

understanding inflation), tends to increase with education, raising concerns as the least-educated individuals are likely more vulnerable to increases in food and energy prices. This posed challenges in adjusting aid packages amid government budget constraints, where monetary policy tightening alone may not sufficiently anchor inflation expectations, particularly in households lacking comprehension of inflation and interest rates (Beckmann & Kiesl-Reiter, 2023). Diversification, as a fundamental economic principle used to hedge against uncertainty, logically suggested that both risk aversion and ambiguity aversion would have a positive impact on its value (Berger & Eeckhoudt, 2021). The study by Stitcha and Sekita (2023) revealed that risk diversification is poorly understood, with only 51 % of Japanese adults answering correctly – referred to as problematic given the vital role of risk in financial decisions. The topic is particularly highlighted by recent economic events and underscores the importance of familiarity with "stock" and "stock mutual funds" when making investment decisions.

Several studies involving different subjects (*e.g.*, investors, Americans, and individuals) proved that financial literacy influences one's financial planning. Johri et al. (2023) explained that financial literacy primarily influences investors' investment decisions, directly impacting their financial planning. It was suggested that investors perceive an enhancement in their awareness of financial objectives, such as retirement planning, with increased financial literacy. Consequently, the 2021 NFCS results concluded that higher financial literacy leveled individuals are more likely to plan for retirement, remain financially resilient, and manage debt better (Lusardi & Streeter, 2023). Augustin and Martin (2022) also found that individuals with higher levels of financial literacy were more inclined towards comprehensive financial planning; and as financial literacy increased, there was a decreased likelihood of having no financial plans. Grounded by this, the current study hypothesized that:

H1. Financial literacy has a direct effect on financial planning.

Financial planning is the process of assessing the financial goals of an individual, taking an inventory of the money and other assets that the person has, determining life goals, and then taking necessary steps to achieve goals in the stipulated period. Meanwhile, the Certified Financial Planner (CFP) Board defined financial planning as a collaborative process that improves one's potential for meeting life goals through financial advice that integrates relevant elements of one's personal and financial circumstances (CFP Board, 2020). Financial literacy is correlated with planning for the future, as the financially literate are more likely to save and plan for retirement (Lusardi et al., 2020). In context, financial decisions are complex, and some level of financial planning can help individuals navigate these complex options. As people become financially literate, they should become more aware of the complexity of their financial decisions, and as a result, consume more comprehensive levels of financial planning (Augustin & Martin, 2022). In order to lessen the risk of financial problems, it is essential to have a solid understanding of financial planning. Therefore, financial literacy is essential in this regard. A financially literate individual will have more significant financial planning than an individual with poor knowledge about financial products and investment markets. Thus, financial literacy is an essential aspect of financial planning (Johri et al., 2023).

Financial goal setting plays a critical role in personal financial planning (Ang, 2024). When individuals set clear, specific, and challenging financial goals, they can be motivated to save and invest more effectively, resulting in better financial outcomes over time (Ang, 2024). Having clear goals can motivate one to plan activities and develop strategies to coordinate work in a way that favors achieving goals (Pe-Than et al., 2020). Locke and Latham (2019) stated that goal setting theory (GST) has been the most prominent theoretical framework for goal setting interventions. First, goal setting directs individuals to focus their efforts on goal-related actions and ignore irrelevant activities. Second, goal setting energizes individuals, allowing them to invest effort in goal pursuit. Third, goals impact persistence, whereby more difficult goals result in a higher effort being invested. Finally, pursuing goals facilitates the discovery and development of task-relevant strategies (Jeong et al., 2023). The simple act of setting a goal was shown to be significant in improving financial outcomes, independent of other interventions (Hackett & Watson, 2022).

In this study, four facets of goal setting were utilized: goal commitment, goal specificity, goal acceptance, and goal difficulty. Goal commitment has been successful in showing a positive influence on various subjects (Gyepi-Garbrah et al., 2023; Hwang et al., 2020). This was also evident with goal specificity, wherein significant contributions on motivation and other concepts such as goal accomplishment perceptions (Gunasti & Ozcan, 2019), as well as leadership and empowerment (Jong & Faerman, 2020) were seen. In a contemporary business context, aligning individual personality and goals through goal acceptance gives an individual a sense of belonging and commitment (Gkizani & Galanakis, 2022). Finally, previous studies have shown a positive linear relationship between goal difficulty and performance levels because people adjust their efforts according to different goals (Groening & Binnewies, 2019).

Xiao and O'Neil (2018a) found that individuals who demonstrate a willingness to establish and work towards long-term goals differ from others in terms of gender, race, age, marital status, presence of children, employment status, educational background, and income. A possible explanation for the link between income and financial literacy is that individuals with higher incomes tend to have more disposable funds, which might motivate them to seek out investment opportunities and acquire more information (Kumar et al., 2019). Peetz et al. (2021) found that income volatility negatively impacts how individuals decide their financial plans, as indicated by their studies where those with more fluctuating incomes tended to overlook financial contingencies (Study 1), and participants receiving volatile payments in a simulated work setting were less inclined to save for a larger delayed payout (Study 2). Similarly, Fulford (2018) found that households with lower incomes tend to aim for a larger emergency savings fund relative to their income, showing a relatively higher level of precautionary behavior compared to higher-income families. Moreover, consumers with characteristics of having a higher income were more likely to set long term goals, thereby exhibiting financial planning (Xiao & O'Neil, 2018a, 2018b). Furthermore, Wann and Burke-Smalley's (2021) study results showed that the odds of having higher levels of financial planning were significantly higher for individuals in the high-income bracket. With that, the current study hypothesized that:

H2. Monthly Income has a direct effect on financial planning.

Occupational status, which is closely linked with income, is one of the most influential factors in saving and investment (Furnham & Cheng, 2019). The responsibility for saving and investing has increasingly fallen on individuals due to actions by governments and employers (Bottazzi & Lusardi, 2021). Abidin et al. (2023) found that the majority of participants, particularly women caring for their families, often resort to working multiple jobs to sustain their savings or investments, highlighting the resilience required in such situations – another factor aside from monthly income. A study conducted in China revealed that households employed in the public sector have stronger saving motives, particularly for education and leaving inheritances to children alongside a higher saving capacity due to better social security. Thus, this significantly influences households' saving decisions with varying saving rates across different occupational types (Xu & Steiner, 2022). Therefore, this study hypothesized that:

H3. Occupation has a direct effect on financial planning.

Most participants in the study of Abidin et al. (2023) shared experiences of financial hardship resulting from insufficient savings, especially during emergencies, emphasizing the importance of not only daily survival but also preparation for unforeseen expenses that could harm families financially. Furthermore, in developed economies, the income

of the household is reported most often as an important variable in influencing financial well-being, and that desired savings and wealth can satisfy safety needs for security to cover future expenses (Kiymaza & Öztürkkal, 2019). On the other hand, Beno (2021) revealed that many employees are expected to realize the potential savings of working from home, attributing this to reduced expenses like transportation and food costs compared to commuting to work. Moreover, Despard et al. (2020) found that the ability to cover expenses alone explains seven to 9 % of the variation in having an emergency fund, indicating that households are less likely to save for emergencies if their income does not occasionally exceed usual expenses, limiting the capacity to build the necessary slack to build assets. Thus, the current study hypothesized that:

H4. Monthly expenses have a direct effect on financial planning.

3. Methodology

3.1. Participants

This study evaluated the financial literacy and financial planning of Filipino millennials. A non-probability purposive sampling method was employed to gather data from 400 respondents through a combination of online surveys administered *via* Google Forms and in-person surveys. The data collection for the survey was conducted over a two-month period, from February 03 to April 30, 2024. The sample size was calculated using the Yamane Taro Formula (Yamane, 1967). Considering a target population of 629,000 Filipino millennials born between 1981 and 1995 (Philippine Statistics Authority, 2023; Prakash & Tiwari, 2021), a sample size calculation for a 95 % confidence level yielded a value of 400 respondents. This study was approved by the university ethics committee before dissemination (FM-RC-22-01-01) and written consent were obtained among participants (FM-RC-22-02-01).

Presented in Table 1 are the descriptive statistics of the demographic profile. It was seen that 53 % are female and 47 % are male, with ages ranging from twenty-eight (28) to forty-three (43) years old. The majority of respondents (71.3 %) were employed full-time, followed by self-employed (14 %), part-time (11.5 %), unemployed (1 %) and students (1 %). Regarding the highest education level attained by the respondents, the majority (67.5 %) held a bachelor's degree. A smaller proportion (13.8 %) had attended some college, while 10.8 % possessed a master's degree. 7.8 % of the respondents reported having completed high school as their highest level of education. Regarding the monthly income of the respondents, the most common income range was ₱10,001 to ₱30,000, accounting for 38.5 % of the sample. The next most frequent income range was ₱30,001 to ₱70,000, representing 37.5 % of the respondents. A smaller proportion (10.5 %) reported a monthly income of less than ₱10,000. Income levels between ₱70,001 and ₱100,000 were reported by 9.8 % of the participants, while only 3.7 % earned over ₱100,000 per month.

Table 1
Respondents' demographic profile ($n = 400$).

Characteristics	Category	N	%
Gender	Female	212	53 %
	Male	188	47 %
	Total	*400*	*100 %*
Age	28–35 years old	186	46.5 %
	35–40 years old	212	53.0 %
	41–55 years old	2	0.50 %
	55 years old and above	0	0 %
	Total	*400*	*100 %*
Employment	Employed full time	285	71.3 %
	Employed part time	46	11.5 %
	Self-employed	56	14 %
	Unemployed	4	1 %
	Student/intern	4	1 %
	Retired	–	–
	Homemaker	5	1.2 %
	Total	*400*	*100 %*
Highest education level	Elementary	–	–
	High school	31	7.8 %
	Some college	55	13.8 %
	Bachelor's degree	271	67.6 %
	Master's degree	43	10.8 %
	Doctoral degree	–	–
	No formal education	–	–
	Total	*400*	*100 %*
Total income (monthly)	Less than 10,000	42	10.5 %
	10,001–30,000	154	38.5 %
	30,001–70,000	150	37.5 %
	70,001–100,000	39	9.8 %
	Above 100,000	15	3.7 %
	Total	*400*	*100 %*
Area of residence	Bangkal	31	7.8 %
	Bel-air	28	7 %
	Carmona	17	4.3 %
	Dasmariñas	12	3 %
	Forbes Park	9	2.2 %
	Kasilawan	14	3.5 %
	La Paz	5	1.3 %
	Magallanes	12	3 %
	Olympia	14	3.5 %
	Palanan	23	5.8 %
	Pio del Pilar	35	8.8 %
	Poblacion	39	9.8 %
	San Antonio	20	5 %
	San Isidro	16	4 %
	San Lorenzo	12	3 %
	Singkamas	6	1.5 %
	Santa Cruz	44	11 %
	Tejeros	13	3.2 %
	Urdaneta	13	3.2 %
	Valenzuela	8	2 %
	Guadalupe Nuevo	10	2.5 %
	Guadalupe Vejo	6	1.5 %
	Pinagkaisahan	13	3.2 %
	Total	*400*	*100 %*

3.2. Measure items

The survey was structured into four sections, following the conceptual framework of the study. The first section is Demographics, which includes factors such as age, gender, place of residence, highest educational attainment, employment status, and monthly income. The second section included the Big Three questions from Lusardi and Mitchell (2011) to assess financial literacy, covering topics such as interest compounding, inflation, and risk diversification. Employing a 5-point Likert Scale (5 = strongly agree, 1 = strongly disagree), the third section delved into Demographic Factors, subdivided into categories such as Monthly Income, Occupation, and Monthly Expenses. Conversely, the fourth section explored Financial Planning within the framework of the Goal Setting Theory, examining aspects like Goal Commitment, Goal Specificity, Goal Acceptance, and Goal Difficulty. All sections and measurement items were a mix of group-formulated questions and items adapted from related literatures, which were assessed by a psychometrician and underwent a pre-test. From a total of 50 initial respondents, an initial output of 0.90 Cronbach's alpha was calculated after minor revisions on the clarity and grammar on the measure items detailed in the Supplementary material.

3.3. Structural equation modeling

The study utilized structural equation modeling (SEM) to better understand the financial stability of Filipino millennials through its variables. SEM is best employed to interpret multiple statistical relationships simultaneously through visualization and model validation (Dash & Paul, 2021). The primary advantage of SEM is its ability to measure complex model relationships while accounting for measurement error inherent in its indicators (Hair et al., 2021, p. 23), making it

well-suited for this study in analyzing the financial planning practices of Filipino millennials in relation to their financial literacy, with the goal of achieving sustainable financial stability. Furthermore, several studies on financial literacy (*e.g.,* Aryan et al., 2024; Bhargava et al., 2022; Rapina et al., 2023) and finance (*e.g.,* Agyapong & Tweneboah, 2023; Ali et al., 2022) utilized SEM for their data analysis, making it ideal for this study to employ SEM as well. Dash and Paul (2021) stated that if the research objective is prediction and theory development, then the appropriate method is partial least squares (PLS-SEM). Existing research that employed the Goal Setting Theory (Locke & Latham, 1990), like in this study, had yielded positive results, thereby contributing to the theory's validity and ongoing development. PLS-SEM helps researchers estimate relatively complex models with minimal sample requirements (*e.g.,* Statsenko & De Zubielqui, 2020; Yeniaras et al., 2020). Moreover, the combination of PLS-SEM's capability of delivering meaningful results, even on small samples, and the method's out-of-sample predictive power (Chin et al., 2020), makes the method particularly powerful (Hair et al., 2021).

4. Results

The initial SEM used to analyze the data determining financial literacy's direct effect on financial planning is shown in Fig. 2. The model consists of 9 latent variables and 36 indicators. In this study, monthly income and monthly expenses have stronger effects on financial planning compared to occupation, while financial literacy was the least impactful. Coming from the "Big Three" financial literacy questions (Lusardi & Mitchell, 2011), it could be said that financial literacy was

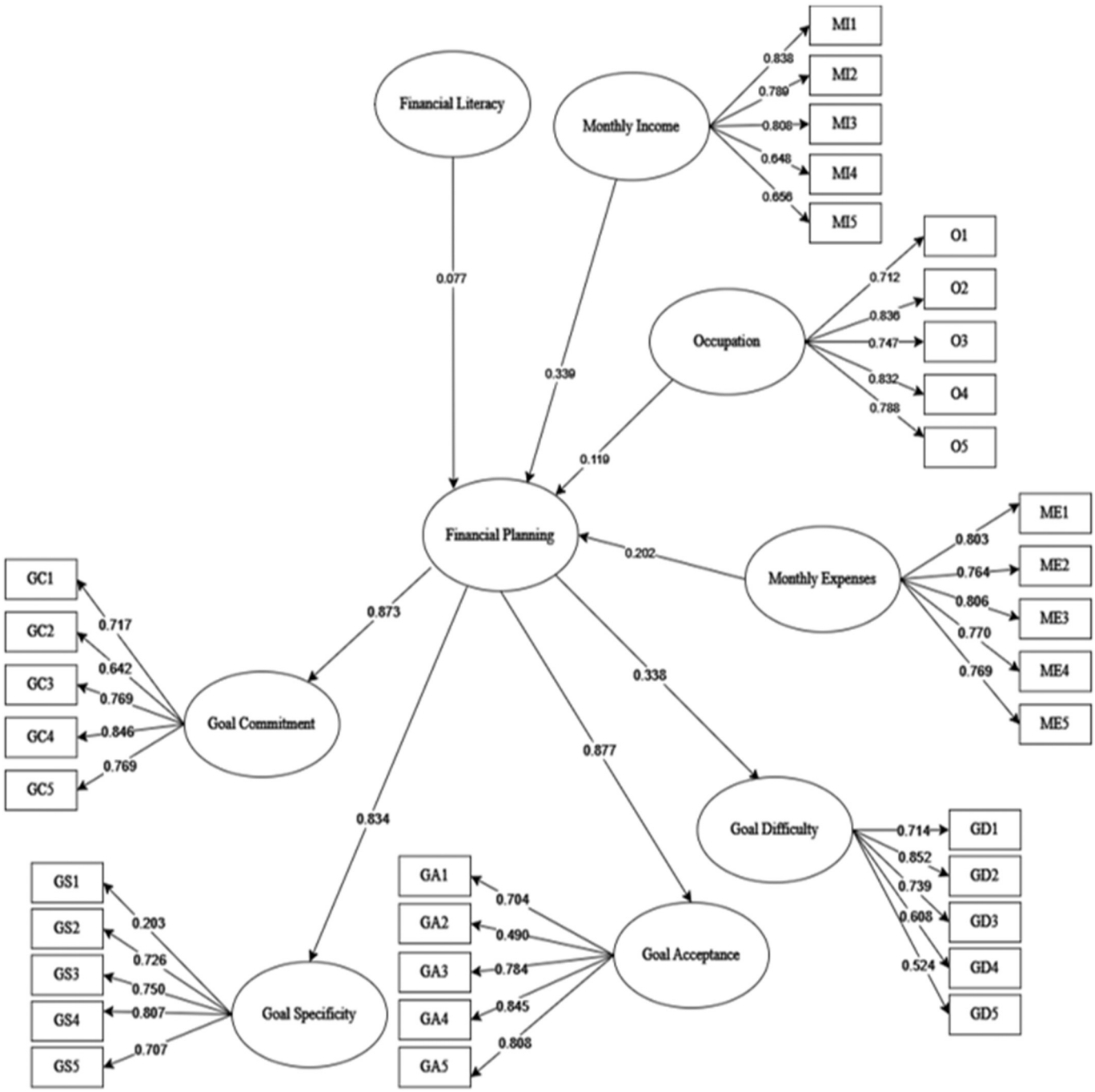

Fig. 2. Initial SEM.

not significantly achieved (p-value <0.05) and has a low beta output of 0.077. From the three vital economic concepts essential for individuals making financial decisions, the mean output and standard deviation as seen in Table 2 implicated that respondents have little knowledge on financial literacy. The output coincides with the findings of Abarcar et al. (2020), wherein information of constraints is present – resulting to low financial literacy among Filipinos. In addition, the Banko Sentral ng Pilipinas reported that only 2 % of Filipino adults correctly answered six financial literacy questions (Mioten, 2024). The explanation represents why financial literacy is an insignificant variable that does not affect financial planning. However, it should be noted that only millennials were considered in this study and may be different if other and older respondents were considered, or if more diverse age group have been collected. Similar to the findings in Japan by Jiang and Shimizu (2024), millennials may not have investment or financial planning behavior despite higher financial literacy. Moreover, the systematic review by Zaimovic et al. (2023) also provided the same evidence among millennials in New Zealand. Reasons may be because of cultural factors, the countries' pension system, or overall behavior of millennials.

In Table 2, the descriptive statistics of the response are shown, alongside the interpretation of data from respondents' and how each variable indicator is being studied, and the calculated average score of the respondents' answers. This points towards expertise in finance being linked to better levels of income and job status and effective expenditure. Following the suggestion of Hair et al. (2021), insignificant latent variables (p-value >0.05) and factor loading less than 0.07 could be removed to represent the final SEM.

Table 2
Statistical analysis of indicators.

Variables	Items	Mean	Standard deviation	Factor loading	
				Initial	Final
Monthly income	MI1	3.596	1.226	0.838	0.877
	MI2	3.653	1.108	0.789	0.850
	MI3	3.764	1.038	0.808	0.842
	MI4	4.185	0.914	0.648	–
	MI5	3.212	1.292	0.656	–
Occupation	O1	3.227	1.223	0.690	–
	O2	3.650	1.123	0.847	0.842
	O3	3.591	1.069	0.753	0.776
	O4	3.512	1.078	0.833	0.845
	O5	3.845	0.922	0.788	0.802
Monthly expenses	ME1	3.882	0.986	0.803	0.786
	ME2	3.374	1.146	0.764	0.752
	ME3	3.522	1.113	0.806	0.819
	ME4	3.170	1.211	0.770	0.774
	ME5	3.665	1.070	0.769	0.784
Goal commitment	GC1	4.172	0.800	0.647	–
	GC2	3.850	0.997	0.490	–
	GC3	4.197	0.804	0.777	0.769
	GC4	4.064	0.854	0.906	0.929
	GC5	4.079	0.841	0.821	0.850
Goal specificity	GS1	3.709	1.089	0.203	–
	GS2	3.833	0.843	0.726	0.712
	GS3	3.874	0.860	0.750	0.758
	GS4	3.941	0.908	0.807	0.826
	GS5	3.872	0.901	0.707	0.714
Goal acceptance	GA1	4.126	0.900	0.690	–
	GA2	3.140	1.347	0.490	–
	GA3	3.973	0.798	0.781	0.809
	GA4	4.022	0.852	0.869	0.887
	GA5	4.076	0.863	0.839	0.851
Goal difficulty	GD1	3.906	1.081	0.714	0.709
	GD2	3.677	1.058	0.852	0.912
	GD3	3.424	1.245	0.739	0.701
	GD4	3.704	1.051	0.608	–
	GD5	3.200	1.180	0.524	–
Financial literacy	Int. Comp.	0.545	0.499	Note: Results are of correct answers by respondents	
	Inflation	0.388	0.488		
	Risk D.	0.338	0.473		

The output implicated that about 0.545 mean score answered correctly on interest compounding, 0.388 regarding inflation, and 0.338 on risk diversification. Moreover, their standard deviations are 0.499, 0.488, and 0.473, respectively.

Furthermore, the convergent validity presented in Table 3 indicates that all the Cronbach's alpha (CA) coefficients exceed the suggested threshold of 0.700, and the composite reliability (CR) meets the acceptance value. In addition, the average variance explained (AVE) for all variables was higher than the 0.40–0.50 threshold recommended by Hair et al. (2010), ranging from 0.477 to 0.734, thus confirming the reliability and convergent validity of the model's latent variables (Dash & Paul, 2021; Hair et al., 2021).

To establish discriminant validity, the Fornell-Lacker criterion (FLC) and heterotrait-monotrait correlation ratio (HTMT) have been employed for assessing convergent validity to be established, which can be seen in Tables 4 and 5. According to Mohd Dzin and Lay (2021), the recommended value using HTMT should be lower than 0.85 or 0.9 to meet the requirement of discriminant validity for the measurement model. Common understanding is that the square root of the AVE (diagonal values) should be higher than the correlational values for FLC to be achieved (Hair et al., 2022). As evidenced by the study of Henseler et al. (2015), when FLC is not achieved in one variable (Financial Planning in this study), it does not automatically mean that discriminant validity has not been achieved. Hair et al. (2022) further emphasized that it may not at all be fatal for FLC not being completely achieved when (1) the HTMT output is achieved, (2) new model and test constructs are created, and (3) constructs are in its early development and utility where studies are still limited. Thus, it could still be acceptable when constructs achieved the HTMT threshold and that other internal validity tests (convergent validity in Table 3) have been achieved (Hair et al., 2022; Henseler et al., 2015; Rigdon, 2012). Hence, this approach established validity through the values of confidence intervals shown in the tables.

To demonstrate the viability of the suggested model, a model fit evaluation was conducted. The acceptability of the proposed model is demonstrated by Table 6, which shows that all parameter estimates were greater than the minimal threshold value as recommended, demonstrating the validity of the proposed model.

The final SEM model revealed significant insights into the financial behaviors of Filipino millennials, highlighting the importance of financial literacy and financial planning, as well as the relationship between these constructs as summarized in Table 7. The variables GC2, GD4, GD5, GS1, MI4, MI5, O1, GC1, and GA1 were removed from the analysis because their beta coefficients were below 0.70 (Hair et al., 2021). To ensure the significance of the remaining variables, the bootstrapping method was performed to confirm the range of HTMT confidence intervals that has gone through bias-corrected and accelerated confidence intervals.

All higher-order constructs demonstrated significant direct effects with financial planning as evidenced by their respective beta coefficients. Among the higher order constructs, goal acceptance had the

Table 3
Convergent validity.

Variables	Cronbach's alpha	Composite reliability	Average Variance Extracted (AVE)
Financial Planning	0.790	0.756	0.477
Goal Acceptance	0.809	0.886	0.722
Goal Commitment	0.810	0.888	0.726
Goal Difficulty	0.769	0.821	0.609
Goal Specificity	0.746	0.840	0.568
Monthly Expenses	0.844	0.888	0.614
Monthly Income	0.820	0.892	0.734
Occupation	0.834	0.889	0.667

Table 4
Fornell-Lacker criterion.

Variable	FP	GA	GC	GD	GS	ME	MI	O
Financial Planning (FP)	0.691							
Goal Acceptance (GA)	0.824	0.850						
Goal Commitment (GC)	0.809	0.607	0.852					
Goal Difficulty (GD)	0.212	0.186	0.199	0.780				
Goal Specificity (GS)	0.597	0.577	0.503	0.205	0.754			
Monthly Expenses (ME)	0.426	0.450	0.305	0.365	0.453	0.783		
Monthly Income (MI)	0.405	0.410	0.322	0.409	0.426	0.681	0.857	
Occupation (O)	0.382	0.400	0.307	0.586	0.389	0.606	0.527	0.817

Table 5
Heterotrait–Monotrait ratio.

Variable	FP	GA	GC	GD	GS	ME	MI	O
Financial Planning								
Goal Acceptance	0.536							
Goal Commitment	0.497	0.733						
Goal Difficulty	0.255	0.221	0.251					
Goal Specificity	0.771	0.732	0.638	0.245				
Monthly Expenses	0.508	0.534	0.352	0.167	0.556			
Monthly Income	0.497	0.497	0.388	0.153	0.544	0.817		
Occupation	0.466	0.486	0.373	0.141	0.492	0.713	0.635	

Table 6
Model fit.

Goodness of fit measures of the SEM	Saturated model	Threshold	Recommended by
SRMR	0.075	<0.08	Ximénez et al. (2022)
Normed Chi-Square	4.012	<5.00	Hu and Bentler (1999)
NFI	0.836	≥0.80	Bai et al. (2025) and Hooper et al. (2008)

Table 7
Summarized results.

Hypothesis	Relationship	Beta	p-Value	Decision
1	FL → FP	0.077	0.153	Not supported
2	MI → FP	0.180	0.003	Accept
3	OC → FP	0.164	0.004	Accept
4	ME → FP	0.205	0.001	Accept
Higher order	FP → GC	0.809	<0.001	Accept
Higher order	FP → GS	0.597	<0.001	Accept
Higher order	FP → GA	0.924	<0.001	Accept
Higher order	FP → GD	0.212	<0.001	Accept

highest significance on financial planning, indicated by a beta coefficient (β) of 0.924, followed by goal commitment with β as 0.809. Goal specificity also exhibited significance on financial planning, with β as 0.597. Among the higher order constructs, goal difficulty had the lowest significance on financial planning, with β as 0.212. Financial literacy exhibited a negligible $\beta = 0.077$ and a *p*-value of 0.153, suggesting a statistically insignificant effect with financial planning. The lower-order constructs all exhibited significant direct effects with financial planning, also as indicated by their corresponding beta coefficients. Monthly expenses exhibited the most significant relationship with financial planning, as indicated by a beta coefficient of 0.205 and a p-value of 0.001. The relationship of monthly income with financial planning also posits significance, with a beta coefficient of 0.18, and a p-value = 0.003. Among the three lower-order constructs, occupation has the least significant effect on financial planning, with a beta coefficient of 0.164 and a p-value of 0.004. Fig. 3 depicts the final SEM utilized in this study.

5. Discussion

Based on the results, goal acceptance had the most significant direct effect on financial planning ($\beta = 0.924$, $p < 0.001$). This observation shows that respondents who are in pursuit of their financial goals are more likely to exhibit financial planning. The respondents' perception of financial planning as a means to achieve their financial goals highlights its significant impact. According to Verner-Filion et al. (2020), passionate individuals have the capability to display high levels of perseverance of effort and consistency towards their interest, with regard to their personal goals. The explanation of the output suggests that respondents demonstrably exerted efforts in alignment with their intentions, leading to a positive impact on achieving personal financial goals. The confluence of these factors contributes to the demonstrably strong relationship between goal acceptance and financial planning. This, in turn, fosters an individual's steadfast commitment to their personal financial objectives.

Second, a significant direct effect was also observed between goal commitment and financial planning ($\beta = 0.809$, $p \leq 0.001$), indicating that respondents are committed to their financial plans. When pursuing various financial goals like saving, spending, or investing, people encounter numerous options associated with the financial goal (Kim et al., 2023), similar to the findings of this study. A goal that one is not committed to attain will not affect the person's actions, as goal commitment may posit an influence through engagement in planning (Earl & Burbury, 2019; Locke & Latham, 2019). The respondents' willingness to conceptualize and put in the action to achieve their goals demonstrates their commitment, highlighting the significant effect that goal commitment has on financial planning.

Third, goal specificity was proven to have a significant direct effect on financial planning ($\beta = 0.597$, $p \leq 0.001$). The results indicated that respondents establish clear and specific goals when creating their financial plans. Specific goals are associated with higher performance compared to vague goals (Jeong et al., 2023) because they define a desired end-state objective, thereby eliminating ambiguity, and accumulating goal progress. It, in turn, increases motivation for achieving goals (Wallace & Etkin, 2018). The results showed that respondents setting specific financial goals provided clarity and direction, enhancing their ability to create and execute comprehensive financial plans. Hence, the results confirm that goal specificity has a significant direct effect on financial planning.

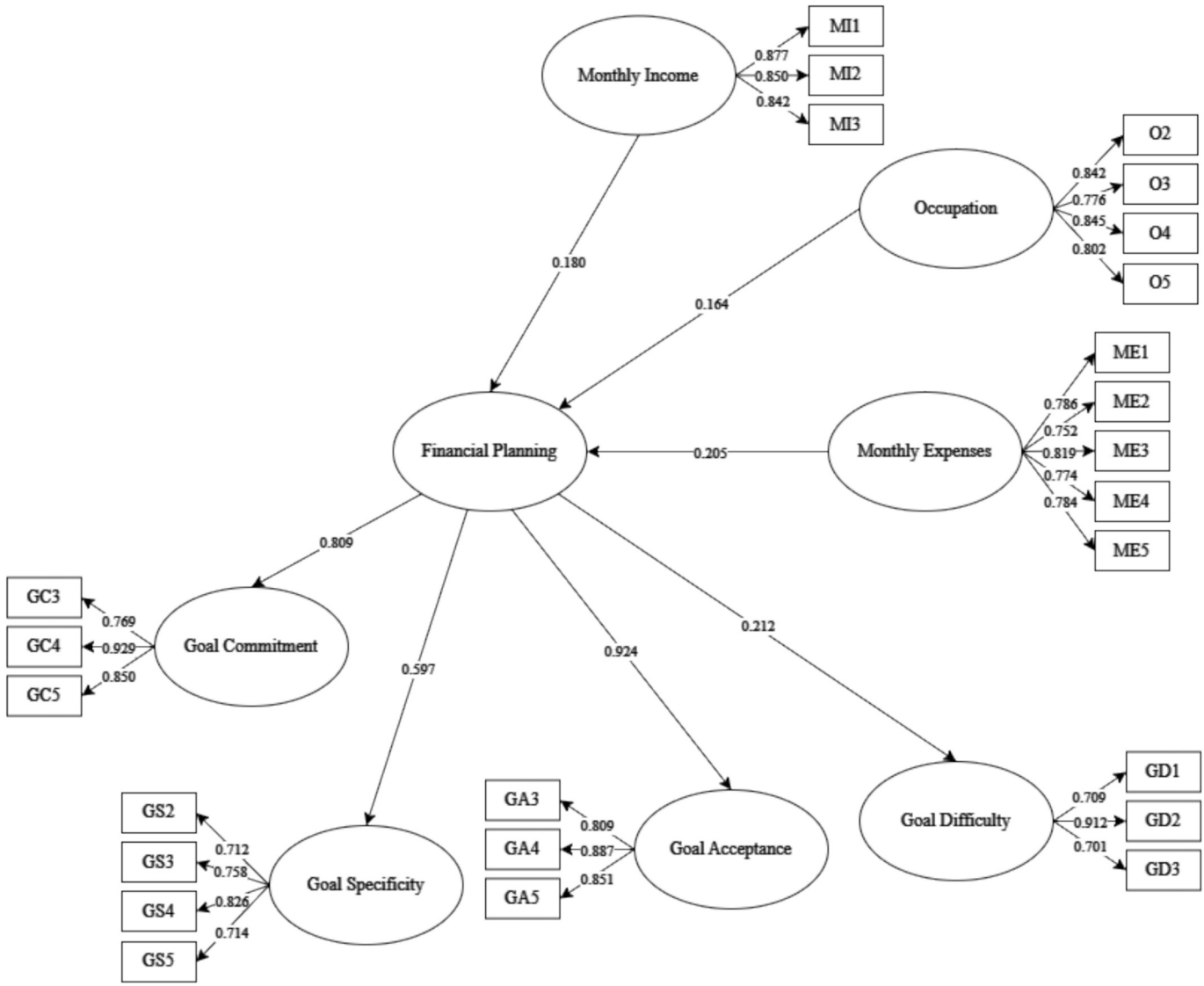

Fig. 3. Final SEM.

Fourth, goal difficulty showed a significant direct effect on financial planning ($\beta = 0.212$, $p \leq 0.001$). This suggests that regardless of respondents' monthly income, occupation, or expenses, they all face challenges that may hinder their ability to achieve financial plans. When setting goals, it is important to ensure that they are attainable because if the difficulty of the future assigned goals is near to being unattainable, then the personal goals will be negatively affected (Fan et al., 2020). Difficult goals can cultivate a sense of accomplishment, thereby stimulating enhanced performance. The enhanced performance is often driven by the effort exerted to achieve these goals, even when they are difficult (Groening & Binnewies, 2019; Yang & Li, 2021). Therefore, it is evident that with sufficient effort, even difficult goals can be attained, and that goal difficulty significantly affects financial planning.

Fifth, it was observed that monthly income has a significant direct effect on financial planning ($\beta = 0.180$, $p = 0.003$). The results showed that respondents take their monthly income into consideration when planning their finances and are mindful of it. This finding aligns with the study by Kumar et al. (2019) which suggests that individuals with higher incomes tend to have more disposable funds available for investment opportunities compared to those with lower incomes. Additionally, higher income enables individuals to set long term goals (Xiao & O'Neil, 2018a). While a higher income is advantageous for investments, a lower income does not necessarily imply a lack of financial planning. Allocation of monthly income to meet financial needs is also a crucial aspect of financial planning, as evidenced by respondents who manage their income to cover their financial needs. Hence, an individual's financial plan, encompassing both needs and investments, is directly influenced by their monthly income.

Sixth, occupation was also shown to have a significant direct effect on financial planning ($\beta = 0.164$, $p = 0.004$). As observed from the results, job security and the type of occupation influence an individual's financial plan. Since occupation is closely linked to income, it can influence savings and investments (Furnham & Cheng, 2019). In accordance, Xiao and O'Neil (2018b) emphasized that budgeting practices hold greater importance for low-income individuals and that financial behaviors vary according to their resources, thereby contributing to financial planning. Given the decline in benefits from both the government and employers, individuals are now more responsible for saving and investing (Bottazzi & Lusardi, 2021). This also aligns with the statement of Xu and Steiner (2022) that different occupational types have varying saving rates. The benefits associated with different occupations can significantly influence financial planning, as those with more benefits allow individuals to allocate more funds towards other needs or additional investments. Thus, it is evident that different

occupations have their own advantages and disadvantages, which directly affect financial planning.

Seventh, monthly expenses demonstrated a significant direct effect on financial planning ($\beta = 0.205$, $p = 0.001$). It is observed that respondents prioritize their monthly expenses when allocating income, which played a significant role in their financial planning. Similar to Despard et al. (2020), a significant portion of an individual's income is typically consumed by expenses, limiting their ability to save or build assets was evident. This also aligns with the study of Peetz and Davydenko (2021), which stated that pursuing goals in the face of temptations, such as necessities and desires, is not easy and people frequently fall short on their goals. High monthly expenses significantly constrain the financial planning capabilities of Filipino households. When faced with substantial recurring costs, their ability to allocate resources towards savings, investments, or other financial plans is markedly limited, thereby hindering their overall financial well-being. Consistent with the findings of Sheehan and Van Ittersum (2018), individuals tend to spend less money once they set budgets or make spending plans before encountering a spending situation. Thus, it is proven that varying monthly expenses directly affect financial planning.

While the results imply that respondents actively engage in financial planning, financial literacy did not show a significant direct effect on their financial planning ($\beta = 0.077$, $p = 0.153$). The results indicated that the majority of respondents had financial literacy scores ranging from 0 to 1, suggesting that Filipino millennials residing in Makati City have relatively low financial literacy. Moreover, the output implied that the driving forces behind their financial planning may be their personal needs and goals rather than financial literacy itself. Furthermore, it appeared that most respondents did not analyze their current financial situation or consulted experts before making financial decisions, which may account for the lack of a significant relationship between financial literacy and financial planning.

Johri et al. (2023), however presented contrary findings. This is because they found that financial literacy influences decisions and directly impacts financial planning. Despite the majority of respondents having a low understanding of basic economic concepts - such as interest compounding, inflation, and risk diversification - they still demonstrated a tendency to save and showed patterns of maintaining a well-managed monthly budget. It also contrasted with the findings of Augustin and Martin (2022), who found that individuals with higher financial literacy are less likely to have no financial plan compared to those with low or no financial literacy. Based on the results, despite showing low financial literacy, the majority of respondents agreed that they engage in financial planning. However, it supports the study of Chen et al. (2024) stating that the escalation of choices through financial planning has not been paralleled by corresponding levels of financial literacy. Concurrently, our study found that variables such as monthly income, occupation, and monthly expenses have a more substantial influence.

5.1. Theoretical implications

This study established the framework of GST by Locke and Latham (1990) as a benchmark for measuring the financial planning of Filipino millennials. Given the significance of the four moderators of GST on financial planning utilized, the theory offered potential applicability for exploring other finance topics such as personal finance, asset/debt management, private equity, investments, financial technology, or cryptocurrency. Moreover, GST can be applied to research on the willingness or intentions of a given population, regardless of the research topic is finance related or not. Millennials, financial institutions, the Philippine government, future researchers, and future generations can make use of the GST to better understand the current landscape of financial planning, empower decision-making, shape relevant policies for financial stability, and inspire new evidence-based research.

5.2. Practical and managerial implications

The analysis of financial literacy and financial planning serves as a foundation to increase the number of individuals making responsible financial decisions, which is particularly significant in developing countries like the Philippines. Encouraging responsible financial behavior can lead to financial inclusion, improved livelihoods, reduced poverty levels, better spending habits, enhanced financial stability, and overall economic growth of a country (Lusardi & Messy, 2023; Omar & Inaba, 2020). Despite years of research highlighting low financial literacy in the Philippines and the BSP's efforts towards financial inclusion, this study emphasizes the need for increased government focus on improving the financial literacy of Filipinos. Collaboration between the government and educational institutions is essential, particularly in developing countries (Runtuwene et al., 2019). Providing free financial education, introducing early financial education, and organizing accessible financial education events nationwide could be effective steps towards achieving the financial literacy establishment goal. These measures can help establish financial planning habits from childhood through young adulthood, leading to improved financial literacy.

Additionally, the government can support financial literacy by promoting job creation through targeted initiatives and partnerships with private organizations. Increased job opportunities can lead to higher incomes and better job security, allowing individuals to save and spend more wisely. Furthermore, private financial organizations are also encouraged to offer more flexible and affordable options for savings and investments to support financial inclusion and encourage widespread engagement in savings. These efforts can improve financial literacy and financial planning, providing a solid knowledge base for Filipinos and facilitating more effective participation in financial inclusion programs and similar initiatives.

5.3. Financial sustainable strategies

Khan, Murshed, et al. (2022) and Khan, Usman, et al., 2022) stated that in today's economy, financial literacy is essential for the whole population and young individuals to sustain well-being. Also, Filipinos should continue to prioritize financial planning to ensure optimal utilization of their funds and achieve long-term financial stability, especially in today's world where financial resources are indispensable for sustenance. The suggested approach supports SDG 1 by ensuring effective use of financial resources to combat poverty.

Considering the empirical evidence indicating low levels of financial literacy among Filipinos, it is imperative to enhance financial literacy to foster greater financial sustainability and understanding of financial products. The approach could align with SDG 4 by emphasizing the need for improved financial education. One effective strategy involves individuals proactively engaged and educated in personal finance, as financial education materials and programs allow people to learn personal finance, with results such as improved investment decisions, with financial education having a strong positive impact on financial literacy, and financial literacy having a positive effect on investment behavior (Kaiser et al., 2022; Kaiser & Menkhoff, 2017; Ullah et al., 2018; Wang et al., 2024).

Through personal finance education, individuals can effectively establish and attain financial objectives while comprehending the advantages and disadvantages of diverse financial instruments that are available in today's world, thereby facilitating informed decision-making. The suggested approach could align with SDG 8 by enhancing financial literacy, which contributes to better financial decision-making and, consequently, supports sustained economic growth and productive employment opportunities.

5.4. Limitations and future direction

Despite its positive outcomes, the current study identifies several

limitations. First, the study focused solely on Makati City, Philippines, which represents only a fraction of the country's population and cannot be considered representative of the entire country. It is recommended to expand the area of the study to include multiple large cities since this study could only reflect respondents in the urban areas of the Philippines. Second, the study's respondents were limited to millennials. Although millennials are recognized as the largest living adult generation (Lusardi, 2019), the results may vary when using a different generation as a sample population, as financial attitudes may differ across generations. It is recommended that future research incorporate other adult generations to broaden the study's scope. Third, the study used only one financial literacy measurement which is "The Big Three" by Lusardi and Mitchell (2011). Since there are various other accepted measures of financial literacy, it is recommended that future researchers incorporate additional measurements alongside "The Big Three" for a more holistic measurement since this study only served as a benchmark. Lastly, the use of a mixed-method research approach is recommended to broaden the effectiveness of using traditional goal-setting theory elements in assessing the financial planning of the respondents. In accordance, other tools aside from SEM could be considered for forecasting sustainable financial behavior and overall goals of respondents. Future researchers could consider the application of machine learning techniques for forecasting behavior, classifying factors, or determining evident score of importance with accuracy rate, even the use of conjoint analysis for financial planning preferences for other strategy build-up to be developed.

6. Conclusion

Financial literacy is crucial as it empowers individuals with the knowledge and skills to save, make sound purchasing decisions, and contribute meaningfully to the nation's economic growth. Considering the limited recent publicly available studies on financial literacy and financial planning in the Philippines, this study narrows the gap by presenting new findings that highlight the low financial literacy in the country and reveal the variables that have a direct effect on financial planning. Emphasizing the need for prioritization on the development of financial literacy and financial planning to ensure the sustained progress of Filipino individuals, communities, and the nation, even though financial literacy alone does not directly affect financial planning.

Employing Goal Setting Theory as the theoretical framework, this study analyzed the direct effect of financial literacy on financial planning. This study, based on a sample of 400 valid responses, employed structural equation modeling (SEM) to assess whether the four goal setting moderators - goal acceptance, goal commitment, goal specificity, and goal difficulty - had significant direct effects on financial planning. The results indicated that all of these goal setting moderators had significant direct effects on financial planning. However, financial literacy did not show a significant direct effect. Additionally, the study found that lower-order constructs, including demographic factors such as monthly income, occupation, and monthly expenses, also had significant direct effects on financial planning.

This study highlights practical implications for policymakers, educational institutions, and financial organizations in the Philippines. To address these needs, it suggests improving financial inclusion programs tailored to the needs of Filipinos. Efforts such as increasing government initiatives to provide free financial education, introducing early financial education programs, and organizing accessible financial education events nationwide are recommended. Additionally, financial organizations should offer more flexible and affordable options for savings and investments to support financial inclusion and encourage widespread engagement in savings. Despite the study's limitations, such as focusing only on Makati City and concentrating on millennials, it can serve as a strong foundation for future research. Expanding the scope to cover a broader geographic area and various age groups can provide a broader perspective on financial literacy and financial planning in the Philippines.

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CRediT authorship contribution statement

Doni Sevaztianne F. Dimaunahan: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Angelique France B. Santiago:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria Christiana C. Eusebio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sumella Maria M. Loteriña:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ardvin Kester S. Ong:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **John Xavier S. Chavez:** Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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